

ADVANCED AND QUANTUM OPTICS — Paper 3

Model Solutions and Marking Scheme

Natural Sciences Tripos Part II — 2026

SECTION A

Question 1 [4 marks] — Focused Gaussian Beam

The Rayleigh range of the incident beam is

$$z_0 = \frac{\pi W_0^2}{\lambda} = \frac{\pi(2.0 \times 10^{-3})^2}{800 \times 10^{-9}} = 15.7 \text{ m}.$$

[0.5 marks]

Since $f = 50 \text{ mm} \ll z_0 = 15.7 \text{ m}$, the beam at the lens has negligible curvature ($R \rightarrow \infty$). In this limit, the lens produces a new waist at approximately its focal plane. The standard result for Gaussian beam focusing gives the new waist radius as

$$W'_0 = \frac{W_0}{\sqrt{1 + (z_0/f)^2}}.$$

When $z_0 \gg f$, this simplifies to

$$W'_0 \approx \frac{W_0 f}{z_0} = \frac{W_0 f}{\pi W_0^2 / \lambda} = \frac{f \lambda}{\pi W_0}.$$

[1.5 marks] for the derivation.

Numerically:

$$W'_0 = \frac{50 \times 10^{-3} \times 800 \times 10^{-9}}{\pi \times 2.0 \times 10^{-3}} = \frac{4.0 \times 10^{-8}}{6.28 \times 10^{-3}} = 6.37 \text{ } \mu\text{m}.$$

[1 mark]

The peak intensity at the new focus is

$$I_0 = \frac{2P}{\pi(W'_0)^2} = \frac{2 \times 5}{\pi \times (6.37 \times 10^{-6})^2} = \frac{10}{1.274 \times 10^{-10}} = 7.85 \times 10^{10} \text{ W/m}^2 \approx 7.9 \text{ GW/m}^2.$$

[1 mark]

Question 2 [4 marks] — Michelson Interferometer Visibility

For a Lorentzian spectrum of FWHM $\Delta\nu$, the complex degree of temporal coherence is

$$g(\tau) = \exp(-\pi \Delta\nu |\tau|) \exp(j2\pi \bar{\nu} \tau).$$

In a Michelson interferometer, the path difference Δ corresponds to a time delay $\tau = \Delta/c$. The fringe visibility is $\mathcal{V} = |g(\tau)|$:

$$\mathcal{V}(\Delta) = \exp\left(-\frac{\pi\Delta\nu}{c}\right) = \exp\left(-\frac{\Delta}{l_c}\right),$$

where the coherence length is $l_c = c/(\pi\Delta\nu)$. [1 mark]

The visibility drops to $1/e$ when $\Delta = l_c$:

$$\Delta_{1/e} = l_c = \frac{c}{\pi\Delta\nu} = \frac{3 \times 10^8}{\pi \times 600 \times 10^6} = 0.159 \text{ m} \approx 16 \text{ cm}.$$

[1 mark]

For a laser with $\Delta\nu_{\text{laser}} = 1 \text{ kHz}$:

$$\Delta_{1/e}^{\text{laser}} = \frac{c}{\pi \times 10^3} = \frac{3 \times 10^8}{3.14 \times 10^3} = 9.55 \times 10^4 \text{ m} \approx 96 \text{ km}.$$

[1 mark]

Comment: The enormous coherence length of a narrow-linewidth laser means that interference fringes of high visibility can be observed over path differences of tens of kilometres. This is exploited in long-baseline interferometry (e.g. LIGO for gravitational-wave detection), precision metrology, and coherent lidar. The thermal source, by contrast, requires path matching to within $\sim 16 \text{ cm}$. [1 mark]

Question 3 [4 marks] — Asymmetric Waveguide Cut-off

For a guided mode, the ray must undergo total internal reflection at *both* interfaces. The more restrictive condition is at the film–substrate boundary (since $n_2 > n_3$), requiring incidence angle θ (measured from the waveguide axis) to satisfy

$$n_1 \sin\left(\frac{\pi}{2} - \theta\right) = n_1 \cos \theta \geq n_2,$$

i.e. $\cos \theta \geq n_2/n_1$, or $\theta \leq \theta_c = \cos^{-1}(n_2/n_1)$. [1 mark]

Why there is a cut-off: In the symmetric waveguide ($n_2 = n_3$), the phase shifts at the two interfaces are identical. The self-consistency condition $2k_y d - 2\phi_r = 2\pi m$ with $m = 0$ always has a solution at any thickness because ϕ_r can accommodate $m = 0$ regardless of d . [1 mark]

In the asymmetric case, the phase shifts $\phi_r^{(\text{sub})}$ and $\phi_r^{(\text{cov})}$ at the two interfaces differ (because $n_2 \neq n_3$). The self-consistency condition becomes

$$2k_y d - \phi_r^{(\text{sub})} - \phi_r^{(\text{cov})} = 2\pi m.$$

At the critical angle $\theta = \theta_c$ (the most grazing guided ray), k_y takes its minimum value and $\phi_r^{(\text{sub})} \rightarrow \pi$. For the $m = 0$ mode to exist, we need

$$2k_y^{\min} d \geq \phi_r^{(\text{sub})}(\theta_c) + \phi_r^{(\text{cov})}(\theta_c).$$

Since $\phi_r^{(\text{cov})}(\theta_c) > 0$ (while it would be $= \phi_r^{(\text{sub})}$ in the symmetric case), the left-hand side must be large enough, which requires d to exceed a minimum value. Below this cut-off thickness, even the fundamental mode cannot satisfy the round-trip phase condition. [2 marks]

Question 4 [4 marks] — Type-I Phase Matching for SHG

In collinear SHG, two fundamental photons at ω produce one at 2ω . Conservation of momentum (phase matching) requires $2k(\omega) = k(2\omega)$, i.e.

$$2 \frac{n(\omega) \omega}{c} = \frac{n(2\omega) 2\omega}{c}, \quad \implies \quad n(\omega) = n(2\omega).$$

[1 mark]

In Type-I, both fundamental photons are ordinary-polarised and the second harmonic is extraordinary. Thus

$$\boxed{n_o(\omega) = n_e(2\omega, \theta_{\text{pm}})}.$$

[1 mark]

Why this is achievable: Normal dispersion means $n(2\omega) > n(\omega)$ for the *same* polarisation. So for ordinary rays, $n_o(2\omega) > n_o(\omega)$ and phase matching is impossible. However, in a negative uniaxial crystal, $n_e < n_o$ for all frequencies. The extraordinary index at 2ω varies with propagation angle θ as

$$\frac{1}{n_e^2(2\omega, \theta)} = \frac{\cos^2 \theta}{n_o^2(2\omega)} + \frac{\sin^2 \theta}{n_e^2(2\omega)}.$$

As θ increases from 0 to 90° , $n_e(2\omega, \theta)$ decreases continuously from $n_o(2\omega)$ (which is too large) down to $n_e(2\omega)$ (which, for a sufficiently birefringent crystal, is less than $n_o(\omega)$). By the intermediate value theorem, there exists an angle θ_{pm} at which $n_e(2\omega, \theta_{\text{pm}}) = n_o(\omega)$. Birefringence thus compensates for normal dispersion. [2 marks]

SECTION B

Question 5 [17 marks] — Optimising Laser Output Power

(a) [3 marks]

In one round trip, the photon-flux density is reduced by mirror reflections and by scattering/absorption in the medium. With $\mathcal{R}_1 = 1$ and $\mathcal{R}_2 = \mathcal{R}$:

$$e^{-2\alpha_r d} = \mathcal{R}_1 \mathcal{R}_2 e^{-2\alpha_s d} = \mathcal{R} e^{-2\alpha_s d}.$$

Taking logarithms:

$$-2\alpha_r d = \ln \mathcal{R} - 2\alpha_s d, \quad \implies \quad \alpha_r = \alpha_s + \frac{1}{2d} \ln \frac{1}{\mathcal{R}}.$$

[1.5 marks]

The threshold condition is $\gamma_0 \ell \geq \alpha_r d$ (gain per pass $\geq$ loss per pass), or equivalently $\gamma_0 \geq \alpha_r$ when $\ell = d$. In general:

$$\gamma_0 \geq \frac{d}{\ell} \left(\alpha_s + \frac{1}{2d} \ln \frac{1}{\mathcal{R}} \right).$$

[1.5 marks]

(b) [4 marks]

Above threshold, gain clamps to the loss: $\gamma_0/(1 + \phi/\phi_s) = \alpha_r$. Solving:

$$\phi = \phi_s \left(\frac{\gamma_0}{\alpha_r} - 1 \right).$$

[1.5 marks]

The output coupler transmits a fraction of the intracavity flux each round trip. The rate at which photons are lost through the output mirror is characterised by the mirror loss coefficient $\alpha_m = \frac{1}{2d} \ln \frac{1}{\mathcal{R}}$. The output flux is the intracavity flux multiplied by the fraction of total loss attributable to the output mirror:

$$\phi_{\text{out}} = \phi \times \frac{\alpha_m}{\alpha_r} \times \alpha_r = \phi \times \alpha_m.$$

More precisely, in one round trip of length $2d$, the fraction of photons leaving through the coupler is $\alpha_m \cdot 2d/(2d) = \alpha_m$ per unit length. The output flux density is thus

$$\phi_{\text{out}} = \phi \cdot \alpha_m = \phi_s \left(\frac{\gamma_0}{\alpha_r} - 1 \right) \frac{1}{2d} \ln \frac{1}{\mathcal{R}}.$$

[2.5 marks]

(c) [5 marks]

With $T = 1 - \mathcal{R} \ll 1$, we use $\ln(1/\mathcal{R}) = -\ln(1 - T) \approx T$. Then $\alpha_m \approx T/(2d)$ and $\alpha_r \approx \alpha_s + T/(2d)$. The output flux becomes

$$\phi_{\text{out}} = \phi_s \left(\frac{\gamma_0}{\alpha_s + T/(2d)} - 1 \right) \frac{T}{2d}.$$

Writing $P_{\text{out}} = h\nu A \phi_{\text{out}} \propto \phi_{\text{out}}$:

$$P_{\text{out}} \propto \frac{T}{2d} \left(\frac{\gamma_0 - \alpha_s - T/(2d)}{\alpha_s + T/(2d)} \right) = \frac{T(\gamma_0 - \alpha_s - T/(2d))}{2d(\alpha_s + T/(2d))}.$$

Dropping the constant $1/(2d)$ and writing $x = T/(2d)$ for clarity:

$$P_{\text{out}} \propto \frac{x(\gamma_0 - \alpha_s - x)}{\alpha_s + x} \equiv f(x).$$

[1.5 marks]

To optimise, differentiate using the quotient rule. Let $u = x(\gamma_0 - \alpha_s - x) = x\gamma_0 - x\alpha_s - x^2$ and $v = \alpha_s + x$:

$$f'(x) = \frac{u'v - uv'}{v^2}, \quad u' = \gamma_0 - \alpha_s - 2x, \quad v' = 1.$$

Setting $f'(x) = 0$: $u'v = uv'$, i.e.

$$(\gamma_0 - \alpha_s - 2x)(\alpha_s + x) = x(\gamma_0 - \alpha_s - x).$$

Expanding the left side:

$$\gamma_0\alpha_s + \gamma_0x - \alpha_s^2 - \alpha_sx - 2\alpha_sx - 2x^2 = x\gamma_0 - x\alpha_s - x^2.$$

$$\gamma_0\alpha_s - \alpha_s^2 + (\gamma_0 - 3\alpha_s)x - 2x^2 = (\gamma_0 - \alpha_s)x - x^2.$$

$$\gamma_0\alpha_s - \alpha_s^2 - 2\alpha_sx - x^2 = 0.$$

$$x^2 + 2\alpha_sx - \alpha_s(\gamma_0 - \alpha_s) = 0.$$

Solving the quadratic (taking the positive root):

$$x = -\alpha_s + \sqrt{\alpha_s^2 + \alpha_s(\gamma_0 - \alpha_s)} = -\alpha_s + \sqrt{\alpha_s\gamma_0}.$$

Since $x = T_{\text{opt}}/(2d)$:

$$\boxed{T_{\text{opt}} = 2d(\sqrt{\gamma_0\alpha_s} - \alpha_s)}.$$

[3.5 marks]

(d) [5 marks]

With $\gamma_0 = 0.50 \text{ m}^{-1}$, $\alpha_s = 0.02 \text{ m}^{-1}$, $d = 0.25 \text{ m}$:

$$\sqrt{\gamma_0\alpha_s} = \sqrt{0.50 \times 0.02} = \sqrt{0.01} = 0.10 \text{ m}^{-1}.$$

$$T_{\text{opt}} = 2 \times 0.25 \times (0.10 - 0.02) = 0.50 \times 0.08 = 0.040 = 4.0\%.$$

[1 mark]

The corresponding loss coefficient is

$$\alpha_r^{\text{opt}} = \alpha_s + \frac{T_{\text{opt}}}{2d} = 0.02 + \frac{0.04}{0.50} = 0.02 + 0.08 = 0.10 \text{ m}^{-1}.$$

This equals $\sqrt{\gamma_0 \alpha_s}$, as expected from the optimisation. The threshold gain is $\gamma_{\text{th}} = \alpha_r^{\text{opt}} = 0.10 \text{ m}^{-1}$. [**1 mark**]

The intracavity flux is $\phi = \phi_s(\gamma_0/\alpha_r - 1) = \phi_s(0.50/0.10 - 1) = 4\phi_s$. The output flux is $\phi_{\text{out}} = \phi \times \alpha_m = 4\phi_s \times 0.08 = 0.32\phi_s$.

The maximum intracavity power (in the limit $\mathcal{R} \rightarrow 1$, $\alpha_r \rightarrow \alpha_s$) would give $\phi_{\text{max}} = \phi_s(\gamma_0/\alpha_s - 1) = \phi_s(25 - 1) = 24\phi_s$. The extraction efficiency is

$$\eta = \frac{P_{\text{out}}}{h\nu A \phi_{\text{max}} c \alpha_m} \approx \frac{\phi_{\text{out}}}{\phi_s \gamma_0} = \frac{0.32}{0.50} = 64\%.$$

More precisely, the fraction of pump power extracted as useful output:

$$\eta_{\text{extraction}} = \frac{\phi_{\text{out}} \times 2d}{\phi_s(\gamma_0 - \alpha_r) \times 2d + \phi_{\text{out}} \times 2d}$$

but the simplest measure is the slope efficiency ratio: $\alpha_m/\alpha_r = 0.08/0.10 = 80\%$ of above-threshold photons exit through the coupler. [**1.5 marks**]

Physical trade-off: If $\mathcal{R}$ is too high (low T), most intracavity photons are wasted on internal losses and very few escape as useful output. If $\mathcal{R}$ is too low (high T), the large mirror loss raises the threshold so much that only a small population inversion excess remains above threshold, reducing the intracavity flux. The optimum balances these: it ensures the laser operates sufficiently far above threshold while extracting the maximum fraction of stimulated photons. [**1.5 marks**]

Question 6 [17 marks] — Self-Focusing and Filamentation

(a) [4 marks]

The intensity profile of a Gaussian beam at its waist is

$$I(r) = I_0 \exp\left(-\frac{2r^2}{w^2}\right), \quad I_0 = \frac{2P}{\pi w^2}.$$

[0.5 marks]

The refractive index is $n(r) = n_0 + n_2 I(r)$. Expanding the Gaussian near the axis ($r \ll w$):

$$n(r) \approx n_0 + n_2 I_0 \left(1 - \frac{2r^2}{w^2}\right).$$

[0.5 marks]

After propagating a distance δz , the phase acquired is $\varphi(r) = (2\pi/\lambda) n(r) \delta z$. The r -dependent (quadratic) part is

$$\Delta\varphi(r) = -\frac{2\pi}{\lambda} n_2 I_0 \delta z \frac{2r^2}{w^2} = -\frac{4\pi n_2 I_0 \delta z}{\lambda w^2} r^2.$$

[1 mark]

A thin lens of focal length f imparts phase $\varphi_{\text{lens}}(r) = -\pi r^2/(\lambda f)$. Equating coefficients of r^2 :

$$\frac{\pi}{\lambda f_{\text{Kerr}}} = \frac{4\pi n_2 I_0 \delta z}{\lambda w^2}, \quad \implies \quad \boxed{f_{\text{Kerr}} = \frac{w^2}{4 n_2 I_0 \delta z}}.$$

[2 marks]

(b) [5 marks]

Diffraction spreading: A collimated Gaussian beam of radius w diverges with half-angle

$$\theta_{\text{diff}} \approx \frac{\lambda}{\pi n_0 w}.$$

At distance δz , the beam radius increases by $\delta w_{\text{diff}} \approx \theta_{\text{diff}} \delta z = \frac{\lambda \delta z}{\pi n_0 w}$. [1 mark]

Kerr focusing: The Kerr lens of focal length f_{Kerr} deflects a ray at radius w by $\delta w_{\text{Kerr}} \approx -w \delta z / f_{\text{Kerr}} = -4n_2 I_0 (\delta z)^2 / w$. But more directly, over a distance δz the beam converges as if focused by a lens at distance f_{Kerr} . For the continuous medium, we require the focusing rate to equal the diffraction rate. The focusing “angle” is

$$\theta_{\text{Kerr}} \approx \left. \frac{w}{f_{\text{Kerr}} / \delta z} \right|_{\text{per unit length}} = \frac{4n_2 I_0}{w} \text{ (per unit length)}.$$

Actually, let us be more careful. Over a length δz , the angular deflection at radius w from the Kerr lens is

$$\delta\theta_{\text{Kerr}} = \frac{w}{f_{\text{Kerr}}} = \frac{4n_2 I_0 \delta z}{w}.$$

[1 mark]

Balance condition: Self-focusing compensates diffraction when $\delta\theta_{\text{Kerr}} = \theta_{\text{diff}}$:

$$\frac{4n_2 I_0 \delta z}{w} = \frac{\lambda \delta z}{\pi n_0 w}.$$

The δz and w cancel:

$$4n_2 I_0 = \frac{\lambda}{\pi n_0}.$$

[1 mark]

Substituting $I_0 = 2P/(\pi w^2)$:

$$4n_2 \frac{2P}{\pi w^2} = \frac{\lambda}{\pi n_0}, \quad \implies \quad P = \frac{\lambda w^2}{8n_0 n_2} \times \frac{\pi}{1}.$$

Wait — the w^2 should not appear in P_{cr} . Let us redo this properly. The balance is

$$4n_2 I_0 = \frac{\lambda}{\pi n_0}, \quad I_0 = \frac{2P_{\text{cr}}}{\pi w^2}.$$

So

$$\frac{8n_2 P_{\text{cr}}}{\pi w^2} = \frac{\lambda}{\pi n_0}, \quad P_{\text{cr}} = \frac{\lambda w^2}{8n_0 n_2}.$$

This still contains w , which is not physical — P_{cr} should be independent of beam size. The resolution is that we should compare the *continuous* self-focusing to diffraction more carefully. Consider a beam segment of length dz as a thin lens of power $1/f_{\text{Kerr}} = 4n_2 I_0 dz/w^2$. For steady self-guiding, the beam must reproduce itself after one “Rayleigh range” $z_0 = \pi n_0 w^2/\lambda$. This requires

$$\frac{z_0}{f_{\text{Kerr}}} \sim 1, \quad \text{i.e.} \quad \frac{\pi n_0 w^2}{\lambda} \cdot \frac{4n_2 I_0}{w^2} \sim 1.$$

[1 mark]

This gives

$$4\pi n_0 n_2 I_0 / \lambda \sim 1, \quad I_0 = \frac{\lambda}{4\pi n_0 n_2}.$$

With $I_0 = 2P_{\text{cr}}/(\pi w^2)$:

$$P_{\text{cr}} = \frac{\pi w^2}{2} \cdot \frac{\lambda}{4\pi n_0 n_2} = \frac{\lambda w^2}{8n_0 n_2}.$$

The apparent w -dependence is an artefact of our simplified model. A rigorous variational treatment (Chiao, Garmire, Townes 1964) replaces the Gaussian profile self-consistently and finds

$$P_{\text{cr}} = \frac{\alpha \lambda^2}{4\pi n_0 n_2},$$

where $\alpha \approx 1.8962$ (the exact numerical factor from solving the eigenvalue problem). In our estimate, comparing with $P_{\text{cr}} \sim \lambda^2/(8\pi n_0 n_2)$ gives $\alpha \sim 0.5$ (order-of-magnitude correct). Accepting $\alpha \approx 1$ for a rough estimate is perfectly fine. [1 mark]

(c) [4 marks]

For $P > P_{\text{cr}}$, the Kerr focusing exceeds diffraction. The beam self-focuses. An estimate of the collapse distance comes from noting that the net focusing power per unit length is

$$\frac{1}{f_{\text{net}}} \sim \frac{1}{z_0} \left(\frac{P}{P_{\text{cr}}} - 1 \right)^{1/2},$$

where we take the excess focusing to scale as $(P/P_{\text{cr}} - 1)^{1/2}$ near threshold (from a perturbative argument). The self-focusing distance is then

$$z_{\text{sf}} \sim z_0 \frac{1}{\sqrt{P/P_{\text{cr}} - 1}} = \frac{\pi n_0 w^2}{\lambda} \frac{1}{\sqrt{P/P_{\text{cr}} - 1}}.$$

This is conventionally written (dropping $\pi/2$ factors) as

$$z_{\text{sf}} \sim \frac{2n_0 w^2}{\lambda} \frac{1}{\sqrt{P/P_{\text{cr}} - 1}}.$$

[2 marks]

Numerical estimates: Using $\alpha \approx 1.9$:

$$P_{\text{cr}} = \frac{1.9 \times (800 \times 10^{-9})^2}{4\pi \times 1.45 \times 3.2 \times 10^{-20}} = \frac{1.9 \times 6.4 \times 10^{-13}}{5.84 \times 10^{-19}} = \frac{1.216 \times 10^{-12}}{5.84 \times 10^{-19}} \approx 2.1 \text{ MW}.$$

[1 mark]

With $P = 10 \text{ MW}$, $P/P_{\text{cr}} \approx 4.8$:

$$z_{\text{sf}} \sim \frac{2 \times 1.45 \times (10^{-3})^2}{800 \times 10^{-9}} \frac{1}{\sqrt{4.8 - 1}} = \frac{2.9 \times 10^{-6}}{8 \times 10^{-7}} \frac{1}{\sqrt{3.8}} = 3.63 \times \frac{1}{1.95} \approx 1.9 \text{ m}.$$

[1 mark]

(d) [4 marks]

Physical explanation: As the beam self-focuses, intensity perturbations across the beam profile are amplified. A local intensity maximum experiences stronger Kerr focusing than its surroundings, concentrating more power and further increasing the local intensity. This is *modulational instability* (also called small-scale self-focusing or beam break-up). The beam fragments into multiple “hot spots”, each of which self-focuses independently as a filament. [2 marks]

Each filament carries approximately the critical power P_{cr} , because a filament with $P < P_{\text{cr}}$ would diffract away, while one with $P \gg P_{\text{cr}}$ would further fragment. The process continues until each fragment carries $\sim P_{\text{cr}}$, establishing a natural “quantum” of self-focused power. [1 mark]

Number of filaments:

$$N \approx \frac{P}{P_{\text{cr}}} = \frac{10 \text{ MW}}{2.1 \text{ MW}} \approx 5 \text{ filaments}.$$

[1 mark]